EXPERIMENT – 32

Implementation of DSA and RSA Signatures to Demonstrate the Effect of Random

CODE:

```
#include <stdio.h>

int power(int a, int b, int m)
{
    int r = 1;
    while (b--)
        r = (r * a) % m;
    return r;
}

int main()
{
    int message = 10;

    /* DSA demonstration: different k values */
    int k1 = 3, k2 = 7;
    int dsa1, dsa2;

    dsa1 = (message * k1) % 23;
    dsa2 = (message * k2) % 23;

    /* RSA demonstration: deterministic signature */
    int d = 7, n = 33;
    int rsa1, rsa2;
```

```
    rsa1 = power(message, d, n);
    rsa2 = power(message, d, n);

    printf("Message = %d\n\n", message);

    printf("DSA Signatures:\n");
    printf("Signature using k = %d : %d\n", k1, dsa1);
    printf("Signature using k = %d : %d\n", k2, dsa2);

    printf("\nRSA Signatures:\n");
    printf("First Signature  : %d\n", rsa1);
    printf("Second Signature : %d\n", rsa2);

    return 0;
}
```

OUTPUT:

```
C:\Users\deept\OneDrive\Doc

Message = 10

DSA Signatures:
Signature using k = 3 : 7
Signature using k = 7 : 1

RSA Signatures:
First Signature  : 10
Second Signature : 10

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Process exited after 0.09057 seconds with return value 0
Press any key to continue . . .
```